

Enhancing precision agriculture through cloud based transformative crop recommendation model

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ABSTRACT ORIGINAL

Modern agriculture relies more on technology to boost food production. It aims to improve both the quality and quantity of food. This paper introduces a novel TCRM (Transformative Crop Recommendation Model). It uses advanced machine learning and cloud platforms to give personalized crop recommendations. Unlike traditional methods, TCRM uses real-time data. It includes environmental and agronomic factors to optimize recommendations. The system has SMS alerts for remote farmers. It outperforms baseline algorithms like Logistic Regression, KNN(k-nearest neighbor), and AdaBoost. TCRM empowers farmers with actionable insights, reducing resource wastage while boosting yield. By offering region-specific recommendations, it enhances profitability and promotes sustainable agricultural practices. The model has 94% accuracy, 94.46% precision, and 94% recall. Its F1 score is 93.97%. The fivefold cross-validation score is 97.67%. These findings show that the model can improve precision farming. It can make agriculture more sustainable and efficient. © The Author(s) 2025.